

Part B Chapter 13b

Agent-based modelling

1 What are agents?

The term agent has been used for a wide variety of applications, ranging from simple batch jobs (termed agent because they can be scheduled in advance to perform tasks on a remote machine) and simple email filters, to mobile applications (termed agent because they can move themselves from computer to computer), to intelligent assistants (because they present themselves to human users as believable characters that manifest intentionality and other aspects of a mental state normally attributed only to humans), to large, open, complex, mission critical systems (such as systems for air traffic control). Some of the key concepts in agent technology lack universally accepted definitions. In particular, there is no real agreement on the definition of an agent. Agents' abilities vary significantly, depending on their roles, capabilities, and their environment. To describe these abilities different notions of agenthood have been introduced. A good overview of the discussions on these themes can be found in (Franklin and Graesser, 1996). The weak notion of agent introduced in (Wooldridge and Jennings, 1995) is often used as a reference (see also (Jennings and Wooldridge, 1998)).

1.1. Weak notion of agency

The weak notion of agency requires the behaviour of agents to exhibit at least the following four types of behaviour:

- Autonomous behaviour
- Responsive behaviour (also called reactive behaviour)
- Pro-active behaviour
- Social behaviour

Autonomy relates to control: although an agent may interact with its environment, the processes performed by an agent are in full control of the agent itself. Jennings and Wooldridge (1998) define autonomous behaviour as:

... the system should be able to act without the direct intervention of humans (or other agents) and should have control over its own actions and internal state.

This means that an agent can only be requested to perform an action, and, as Jennings and Wooldridge (1998) state:

The decision about whether to act upon the request lies with the recipient.

Examples of autonomous processes are: process control systems (like thermostats, missile guiding systems, and nuclear reactor control systems), software daemons (e.g., one that monitors a user's incoming email and attracts attention by displaying an icon when new, incoming email is detected), operating systems.

Many processes that exhibit autonomous behaviour are currently termed agents. However, if such agents do not exhibit flexible behaviour, they are not considered to be intelligent agents. According to (Jennings and Wooldridge, 1998) an intelligent agent is a computer system that is capable of performing flexible autonomous actions to meet its design objectives.

Intelligence requires flexibility with respect to autonomous actions. Flexibility in turn, implies the ability to behave responsively, socially, and pro-actively.

An agent exhibits responsive (or reactive) behaviour if it reacts or responds to new information from its environment. Jennings and Wooldridge (1998) define responsive behaviour as follows:

Agents should perceive their environment (which may be the physical world, a user, a collection of agents, the Internet, etc.) and respond in a timely fashion to changes that occur in it.

A barometer is a simple example of a system that exhibits responsive behaviour: It continually receives new information about the current air pressure and responds to this new information by adjusting its dial.

Pro-active behaviour is defined by (Jennings and Wooldridge, 1998) as follows:

Agents should not simply act in response to their environment, they should be able to exhibit opportunistic, goal-directed behaviour and take the initiative where appropriate.

Pro-active behaviour is the most difficult of the required types of behaviour for an agent defined according to the weak agent notion. For example, pro-active behaviour can occur simultaneously with responsive behaviour. It is possible to respond to incoming new information in an opportunistic manner according to some goals. Also initiatives can be taken in response to incoming new information from the environment, and thus this behaviour resembles responsive behaviour. However, it is also possible to behave pro-actively when no new information is acquired from the environment. This last type of behaviour can by no means be called responsive behaviour. A more elaborate comparison between responsive behaviour and pro-active animal behaviour can be found in (Jonker and Treur, 1998).

An agent exhibits social behaviour if it communicates and co-operates with other agents. Jennings and Wooldridge define social behaviour as follows:

Agents should be able to interact, when they deem appropriate, with other artificial agents and humans in order to complete their own problem solving and to help others with their activities.

An example of an agent that exhibits social behaviour is a car. It communicates with its human user by way of its dials (outgoing communication dials for: speed, fuel, temperature) and its control mechanisms (incoming communication: pedals, steering wheel, and gears). It co-operates with its human user, e.g., by going in the direction indicated by the user, with the speed set by the user.

1.2 Other notions of agency

Agents can also be required to have additional characteristics. In this section three of these characteristics are discussed: adaptivity, pro-creativity, and intentionality.

Adaptivity is a characteristic that is vital in some systems. An adaptive agent learns and its behaviour improves with experience. This behaviour is vital in environments that change over

time in ways that would make a non-adaptive agent obsolete, or have no chance of survival. This characteristic is often modelled in simulations of societies of small agents, but also, for example, in adaptive user interface agents.

Pro-creativity is of similar importance to gradually develop agents that fit certain conditions better. A computer virus is a very infamous form of a pro-creative agent. The chance of survival is often measured in terms of a fitness function. This characteristic is modelled often in simulations of societies of small agents (see the literature in the area of Artificial Life).

According to Dennett (1987) an intentional system is an entity

... whose behaviour can be predicted by the method of attributing beliefs, designs and rational acumen.

Mentalistic and intentional notions such as beliefs, desires, intentions, commitments, goals, plans, preference, choice, awareness, may be assigned to agents. The stronger notion of agenthood in which agents are described in terms of this type of notions provides additional metaphorical support for the design of agents.

2 Agents in model-based intelligent applications

The previous sections explained the notion of agenthood. As shown in the previous chapter domain models can be integrated within agents in four different ways. A domain model can be used directly within an agent in order to simulate human behaviour. However, in this course domain models will mainly be used to develop analysis, support or adaptation models to be used to create intelligent agents. The notion of agenthood is used in this course as a framework for developing ambient agents; autonomous agents capable of analysing human behaviour and pro-actively offering support to satisfy desires about particular concepts.

2.1 Primitive agent concepts

The notions of agenthood, as discussed in Section 1.1 and 1.2 are highly abstract notions. To design agents, it is necessary to be familiar with the primitive agent concepts. These primitive concepts serve as vocabulary to express analyses and designs of agents and multi-agent systems. Two classes of primitive notions are distinguished: those used to describe the behaviour of agents in terms of their external (or public) states and interactions, and those

used to describe the behaviour of agents in terms of their internal (or private) states, and processes.

EXTERNAL PRIMITIVE CONCEPTS

Two types of interaction of an agent with its environment are distinguished, depending on whether the interaction takes place with an agent or another type of entity (called an external world), for example a database, or the material world. For each of these two types of interaction specific terminology is used.

INTERACTION WITH THE EXTERNAL WORLD

Two primitive types of interaction with the external world are distinguished. The first type of interaction, observation, does not change the world state, but does change the information the agent has about the world, whereas the second type, performing an action, does change the world state, but does not change the information the agent has about the world. Combinations of these primitive types of interaction are possible; for example, performing an action, and observing its results.

Observations

Two types of observation (or sensing) can be distinguished:

- **Passive:** The agent passively receives the results of observations without taking any initiative or control to observe.
- **Active:** The agent actively initiates and controls the observations it performs; this enables the agent to focus its observations and limit the amount of information acquired.

Execution of Actions

The agent changes the state of its environment by initiating and executing actions.

COMMUNICATION WITH OTHER AGENTS

Two directions of communication are distinguished:

1. **Outgoing:** The agent is capable of communicating to another agent

2. Incoming: The agent is capable of receiving communication from another agent

Agents are often capable of communication in both directions.

INTERNAL PRIMITIVE CONCEPTS

A description in terms of the external primitive concepts abstracts from what is inside the agent. In addition to descriptions of agents in terms of the external concepts, often descriptions in terms of internal concepts are useful. The following internal primitive agent concepts are distinguished.

WORLD AND AGENT MODELS

An agent may create and maintain information on (a model of) the external world based on its observations of that world, on information about that world communicated by other agents, and its own knowledge about the world. The agent may also create and maintain information on (models of) other agents in its environment based on its observations of these agents as they behave in the external world, on information about these agents communicated by other agents, and its own knowledge about other agents.

SELF MODEL AND HISTORY

Some agents create and maintain information on (a model of) their own characteristics, internal state, and behaviour. An agent may also create and maintain a history of the world model, agent models, self model, or own and group processes.

GOALS AND PLANS

To obtain pro-active, goal-directed behaviour, an agent often represents, generates, and uses explicit goals and its own plans of action in its processing.

2.2 Input for intelligent support agents

For an agent to make use of a domain model, it needs to be aware of its surroundings. This can be accomplished by adding sensors or other methods for the agent to gather information. Choosing a good method for model input, can be very difficult. Even though on first sight some choices may be obvious, there are a number of aspects to consider before choosing one particular method. Without going into much detail, this chapter will provide some pointers to think about when choosing the input for an ambient agent.

2.2.1 Sensors

There are a large number of different sensors available measuring all kinds of things like location, (body-) temperature, (eye-) movement, brain and heart-activity, et cetera. Often, based on these measurements, multiple results can be gathered. For example, an ECG-sensor measuring electrical heart activity can be used to gather heart rate, but is also able to provide a measurement for heart rate variability. This large collection of sensors and measurements often provides several options for an agent to observe the world. Below a number of aspects to consider when choosing a sensor are discussed (based on Both et al., 2011). This list however is by no means complete and many more considerations can be made when picking sensors for your application.

SENSITIVITY

The sensitivity refers to the capability of the measurement to detect differences in the performance of the human. Some measurements might be relatively coarse grained whereas others can measure on a fine granularity. A mobile application requiring knowledge about a user's location could either use one based on cell tower triangulation or gps-signal. While the latter is more sensitive, the former is more often available.

INTRUSIVENESS

One of the criteria related to the sensing devices themselves is the intrusiveness of the sensor to perform measurements. In case the sensors are very intrusive, this might lead to the human feeling uncomfortable, as there is a continuous awareness of everything being measured. An example will be given in the next paragraph.

RELIABILITY

When sensors are used another important aspect is how robust the measurements are. Some measurements might only be robust when they are performed under laboratory conditions whereas the developed assistant agent might be meant for more demanding conditions. There is often a trade-off between the intrusiveness of sensors, and their robustness. Measuring heart rate using electrodes is more robust compared to measuring it via a watch. However, the latter is less intrusive compared to having electrodes attached to your body.

IMPLEMENTATION REQUIREMENTS

Another criterion includes the requirements of the measurement to be performed and how difficult it is for an ambient agent to interpret the sensing data. Some data is very easily understandable (e.g. the heart rate), whereas other measurements require the ambient agent to have a more thorough knowledge on how to use the information (e.g. an EEG).

TASK DEPENDENCE

In the design of an ambient agent, the goal is often not only to investigate support for a single task, but for multiple tasks and to allow for more generic support. Therefore, it is important that the agent does not entirely depend on measurements that highly depend on the characteristics of the tasks being performed. Therefore, the portability of the measurements to other tasks can also be considered when choosing sensing devices.

COST

Another maybe quite obvious factor to consider is cost. Some sensors are relatively cheap, whereas others can be quite expensive. Depending on the application, keeping the costs low might be important.

PRIVACY

The last aspect mentioned is privacy. When taking measurements, it is important to consider potential privacy risks. For example, people might be hesitant to add a webcam to the bedroom for an ambient agent designed to help users sleep better.

2.2.2. Other methods of input

Beside sensing, other options are available for an ambient agent to gather information that may be required to reason within a particular domain. The following paragraphs mention a couple of possibilities.

DEFAULT VALUES

In some cases it might be sufficient to set a default value for a particular aspect without actually measuring it. Such assumptions are of course very prone to errors, but when carefully considered default values can be very useful. Among the points that should be considered when thinking about using such values for your ambient agent are as follows. How large is the chance for errors? In particular applications, the target user group might be very

homogeneous in some particular aspect that a default value could be very accurate most of the time. Related is another point to think about; the impact of a wrong value. When a value is incorrect, does your agent make disastrous errors or only small mistakes? Also, it might be important to consider whether the ambient agent will be able to learn from its mistakes. If overtime errors can help the agent to improve upon those values, mistakes in the beginning could be acceptable. In conclusion, when choosing a default value it is important to consider what happens when those are incorrect.

COMMUNICATION FROM OTHER AGENTS

Another possibility is that your ambient agent will be part of a multi-agent system. In that case, communication from other agents could provide valuable information for your agent, without the need of actually observing, measuring or reasoning about it yourself. In essence, the other agent can be thought of as a sensor and similar considerations as before have to be made. Is the agent reliable, sensitive, intrusive, etcetera.

OTHER SOURCES

Thinking outside the box can also provide interesting sources for information. Not everything is originally designed as a sensor, but can more or less be used as one. As an example, take the computer mouse. At first sight, it might not look like a sensing device. However, for particular applications, measuring mouse movements can be an indication of stress. Think of abrupt or fast movements and overshooting the target as measurements of stress. Albeit not very precise and reliable, the measurement itself is very unintrusive and cheap. As can be seen, the message here is not necessarily that there are other sources besides sensors, but that many sources can be used as sensors.

- the physical and biological environment,
- physiological and cognitive processes within humans,
- behavioural processes of a human in interaction with his or her environment
- social and organisational processes based on interaction between humans.

3 Conclusion

Agent models can be used to describe phenomena in the natural world (such as animals or humans and their behaviour and interaction), or for artificially created (engineered) processes such as, for example:

- software agents
- intelligent cars
- search bots and recommender systems on Internet
- smart homes
- robots

For this course, agent models will be used for both cases: to model (natural) human agents and to model (artificial) ambient agents. Here ‘ambient’ means in the immediate environment of humans that are supported by them, for example, on their cell phone. To model human functioning agent models will be addressed, for example, for humans with behaviour motivated by desires, and humans generating emotions and feelings. Moreover, for a number of applications it will be shown how domain models can be integrated within intelligent agent models in order to give these ambient agents some understanding of the part of reality they are dealing with. Such understanding enables them to perform actions in a more informed, knowledgeable manner, and to show more human-like behaviour in interaction with humans.